Modeling Infectious Diseases Using R: An Introduction

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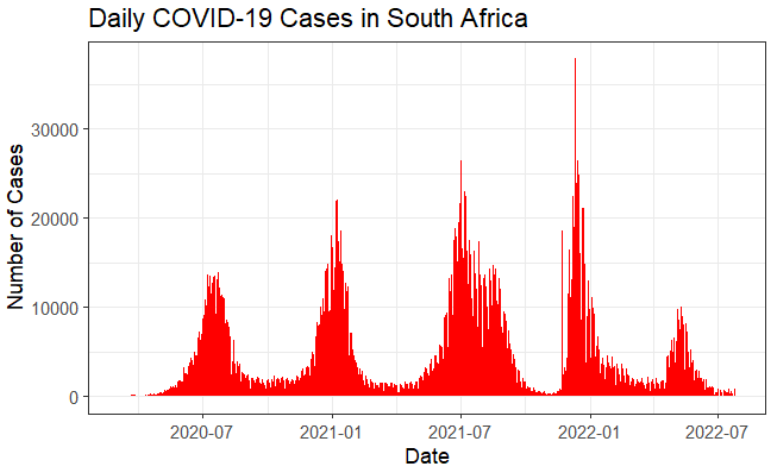
First Training workshop, Cape Town, South Africa
12/10/2026-16/12/2026

STEAM-3D
Sustainable Training in Epidemiological Analysis and Modelling for Data-Driven Decision Making in Africa

SOUTH AFRICA | MALAWI | MOZAMBIQUE | CAMEROON | BELGIUM

Course’s structure

Infectious disease course



- Our course (Tuesday & Wednesday):
  - Focus on understanding how a disease spread in the population.
  - We do not model the data (will do this next workshop in the spring).

What do we cover in the course ?

- Introduction:
  - Example tour and terminology
  - The modeling framework.
  - Transmission and Transmission models.
- The SIR model:
  - Incidence Data & SIR transmission model in R.
  - The SIR model in time homogeneity setting.
  - More about modeling.
- Multiple populations:
  - Understanding contact/mixing patterns in the population.
  - Age structured population.
- Vaccination: Intervention and Control.
- Estimation of Transmission parameters.
- Use R to model infectious diseases transmission.
- Case studies:
  - Transmission model for HIV/AIDS.
  - Transmission models for HCV among injecting drug users.

Software for practical sessions

- For the practical sessions:
  - R package: **`deSolve`**.
  - R programs for practical sessions.
  - R/R Studio.

Course plan

- Tomorrow: transmission models.
- Wednesday: practical sessions:
  - Very shout intro to R (reptation of topics from the R course).
  - Transmission models in R.

Course’s materials

Course materials

Course

Slides for the course (PDF).

Practical sessions

Slides for the practical sessions (PDF).

R programs for the practical sessions (PDF).

Online tutorials

Online tutorials (HTML).

Rmd files for the Online tutorials.

Download the materials before lectures & practical sessions.

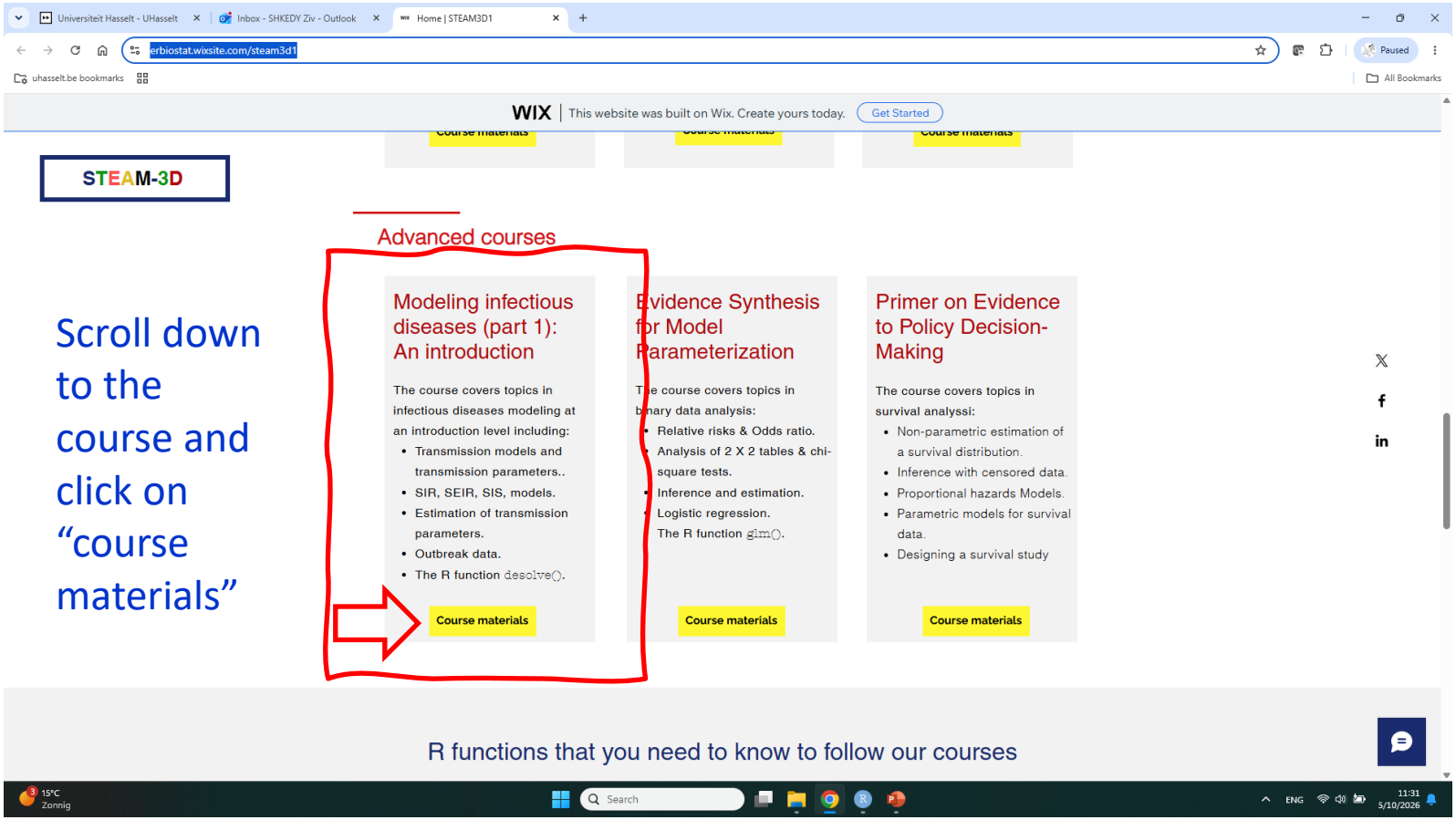
Where you can find the course materials ?

- All course materials are available online.
- The link to the workshop website:

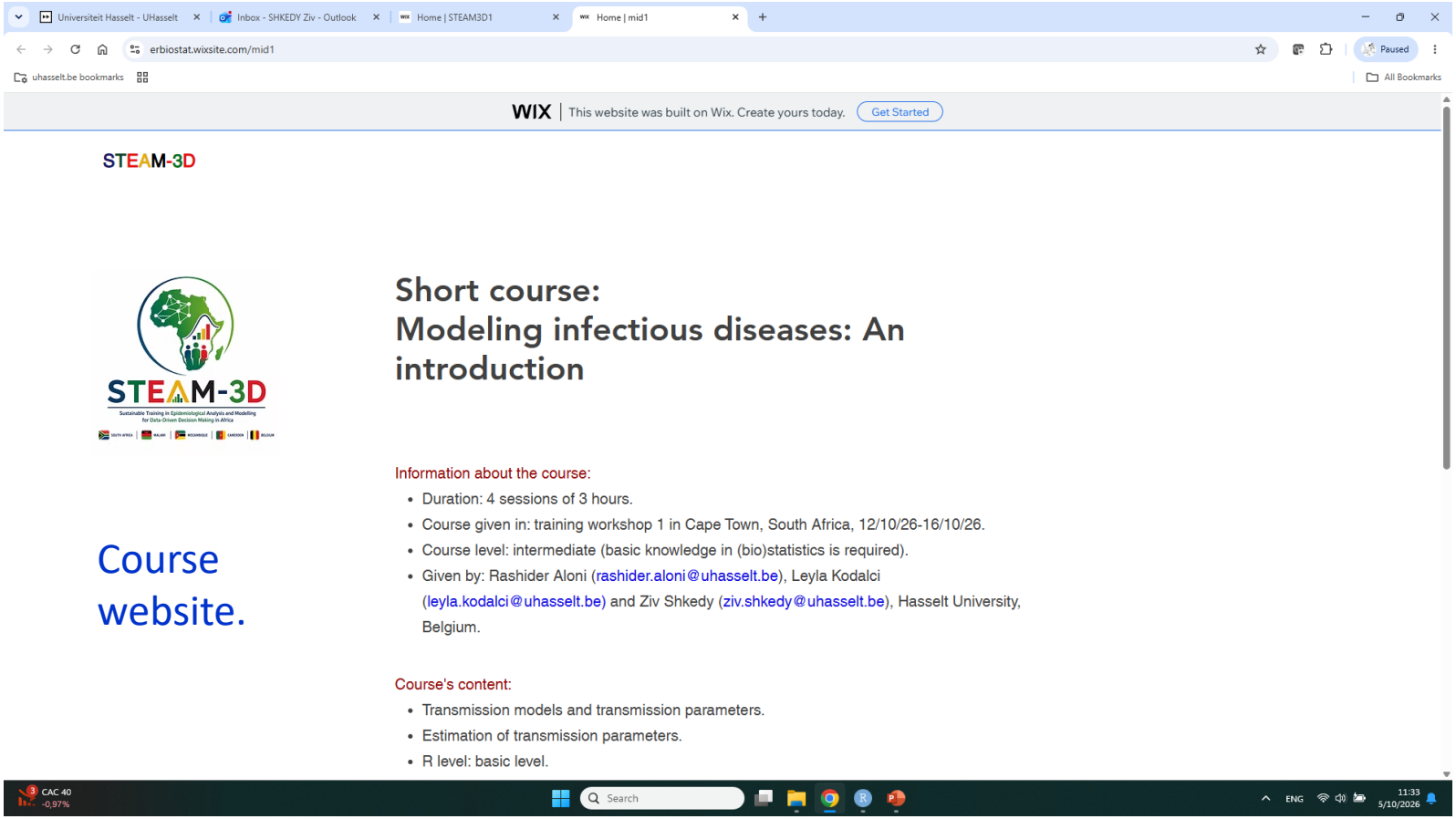


https://erbiostat.wixsite.com/steam3d1

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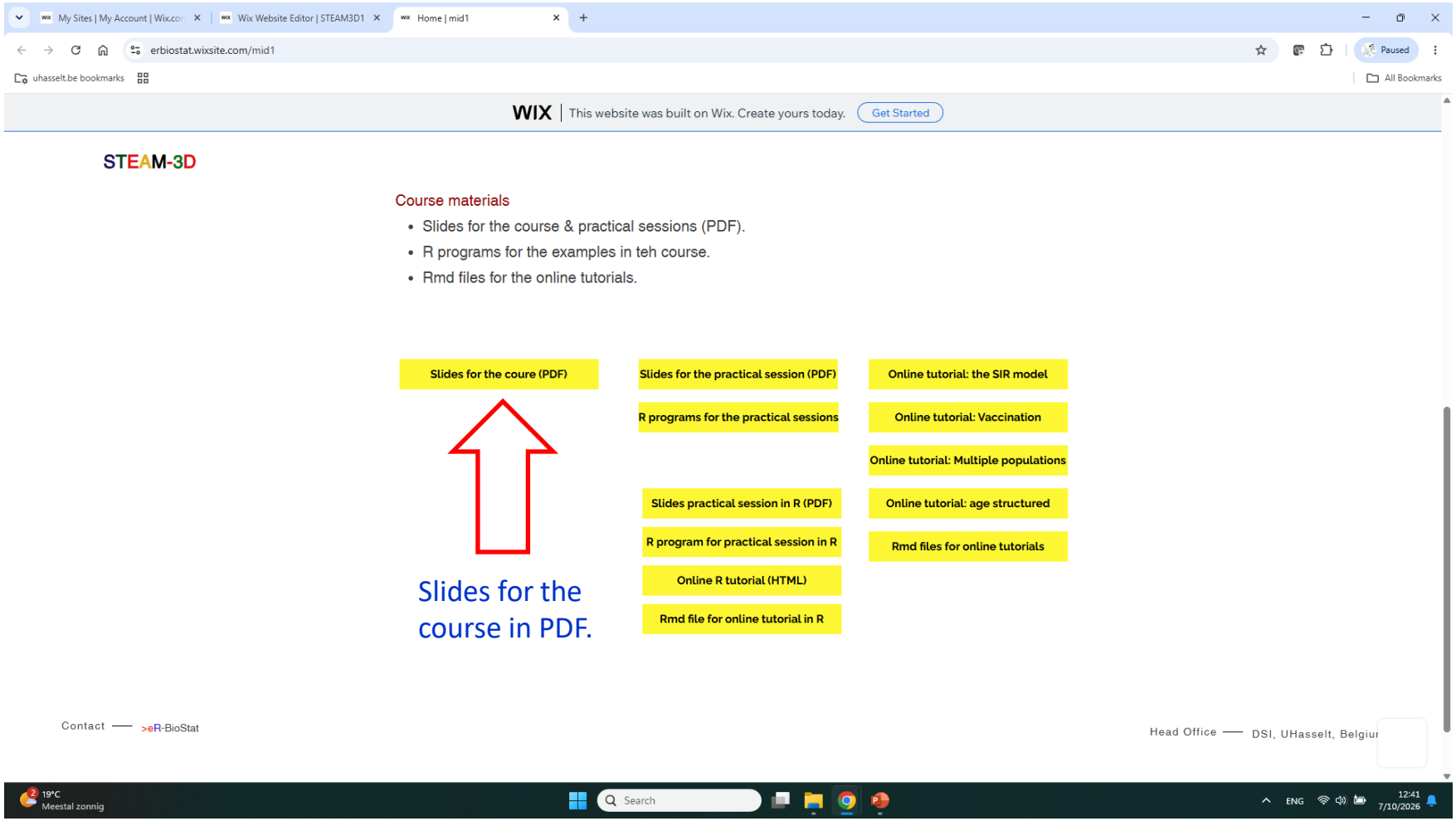
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Course website.

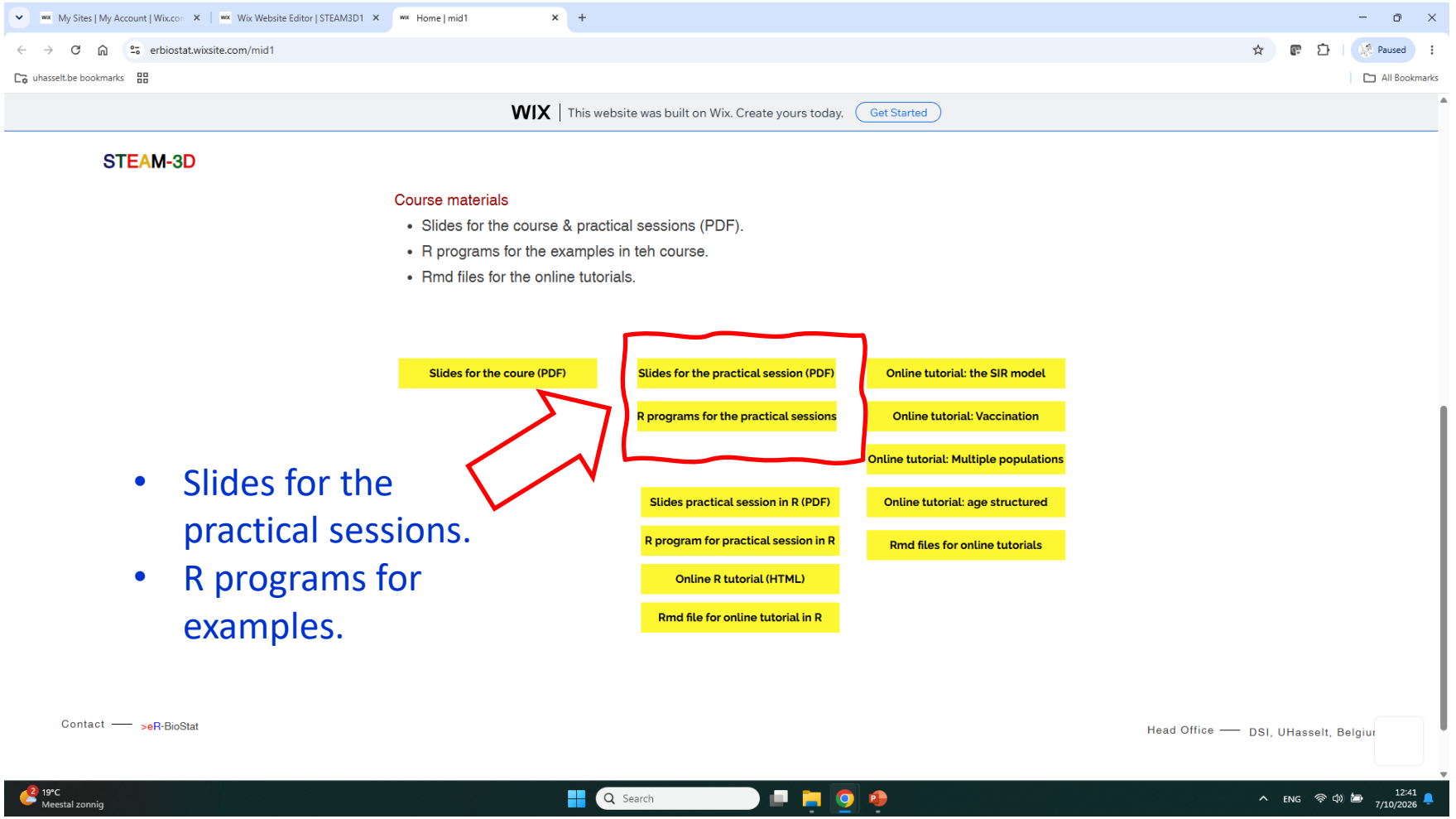
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Slides for the course



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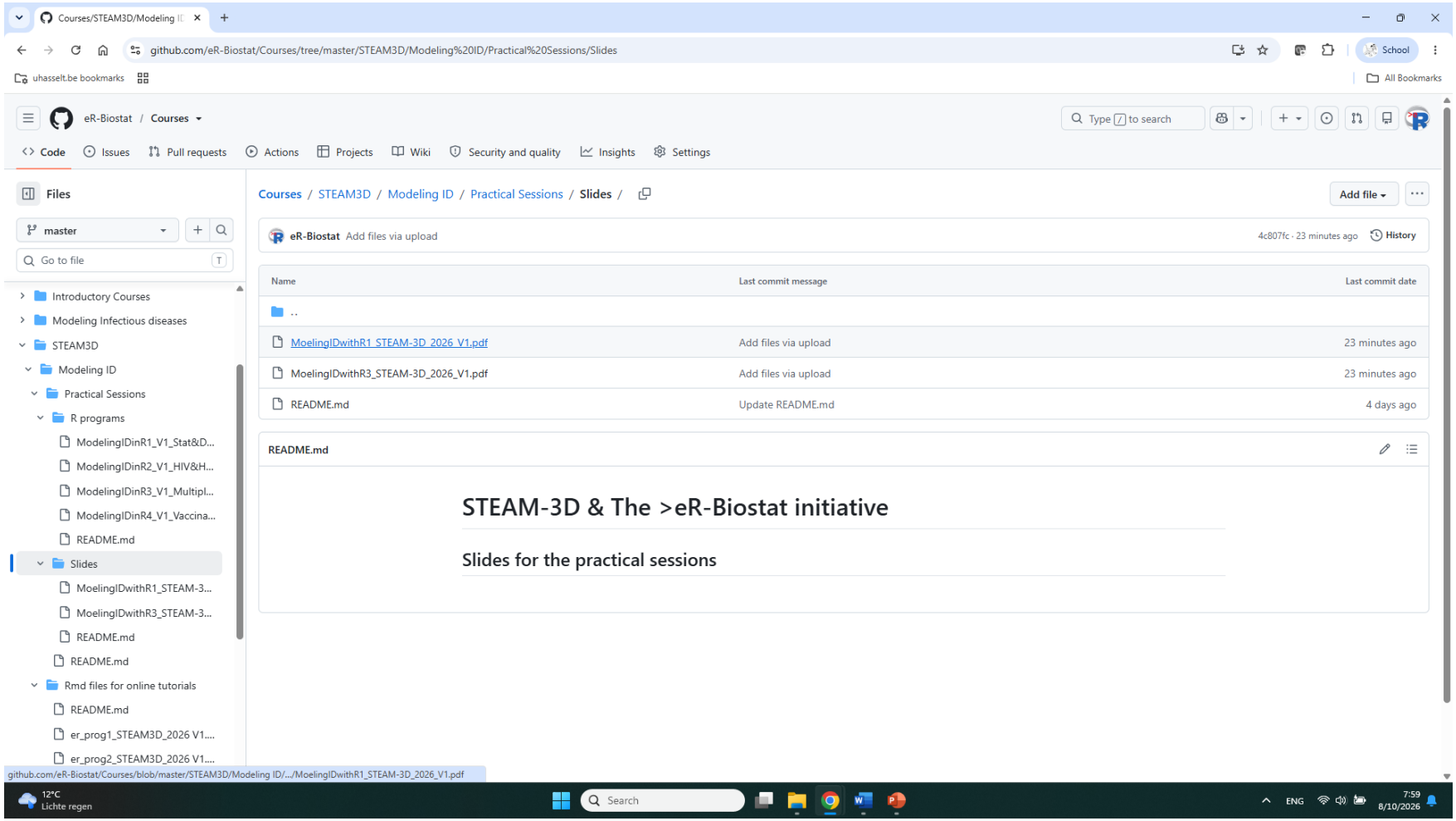
Practical sessions



- Slides for the practical sessions.
- R programs for examples.

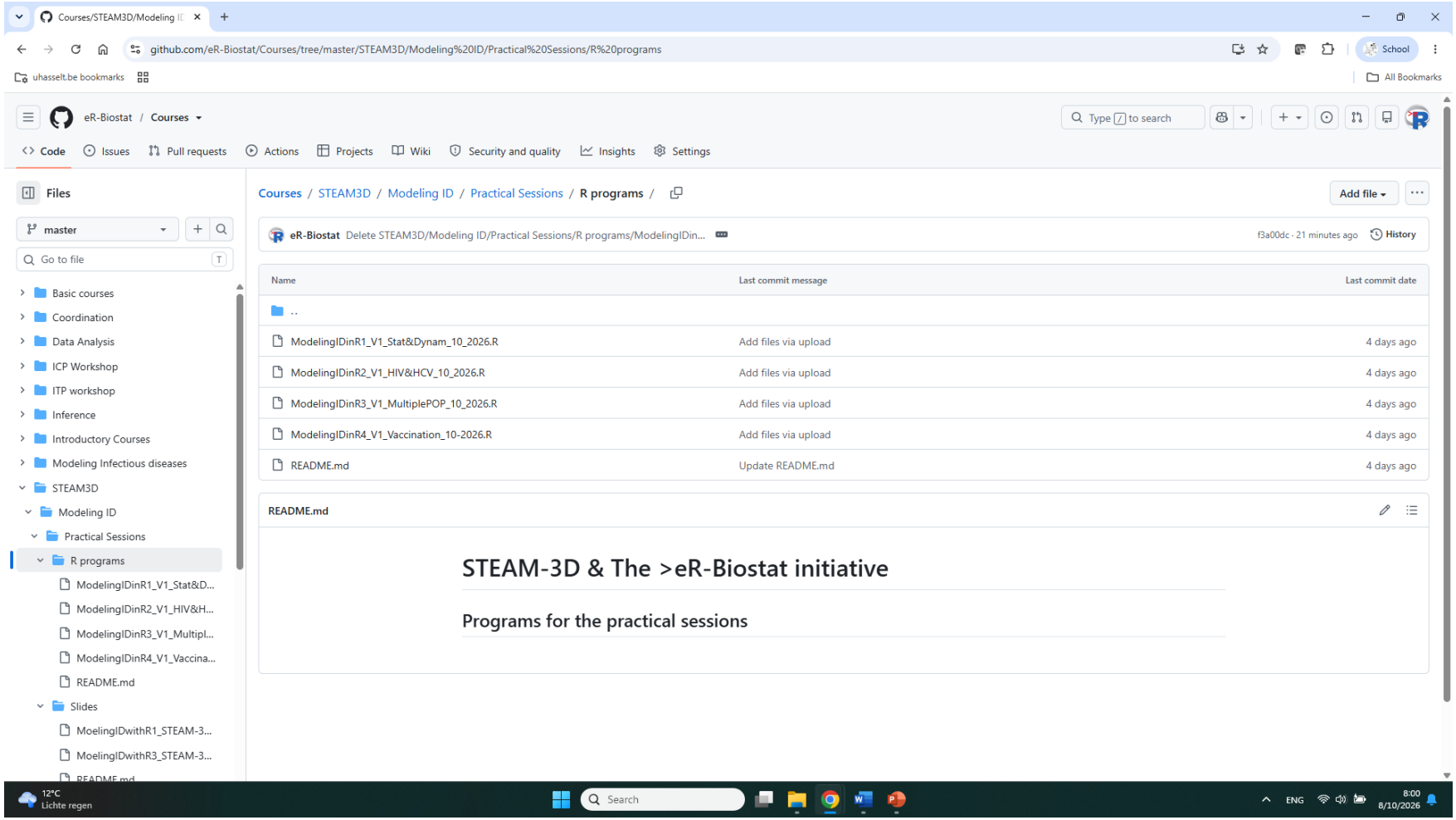
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Practical sessions: slides



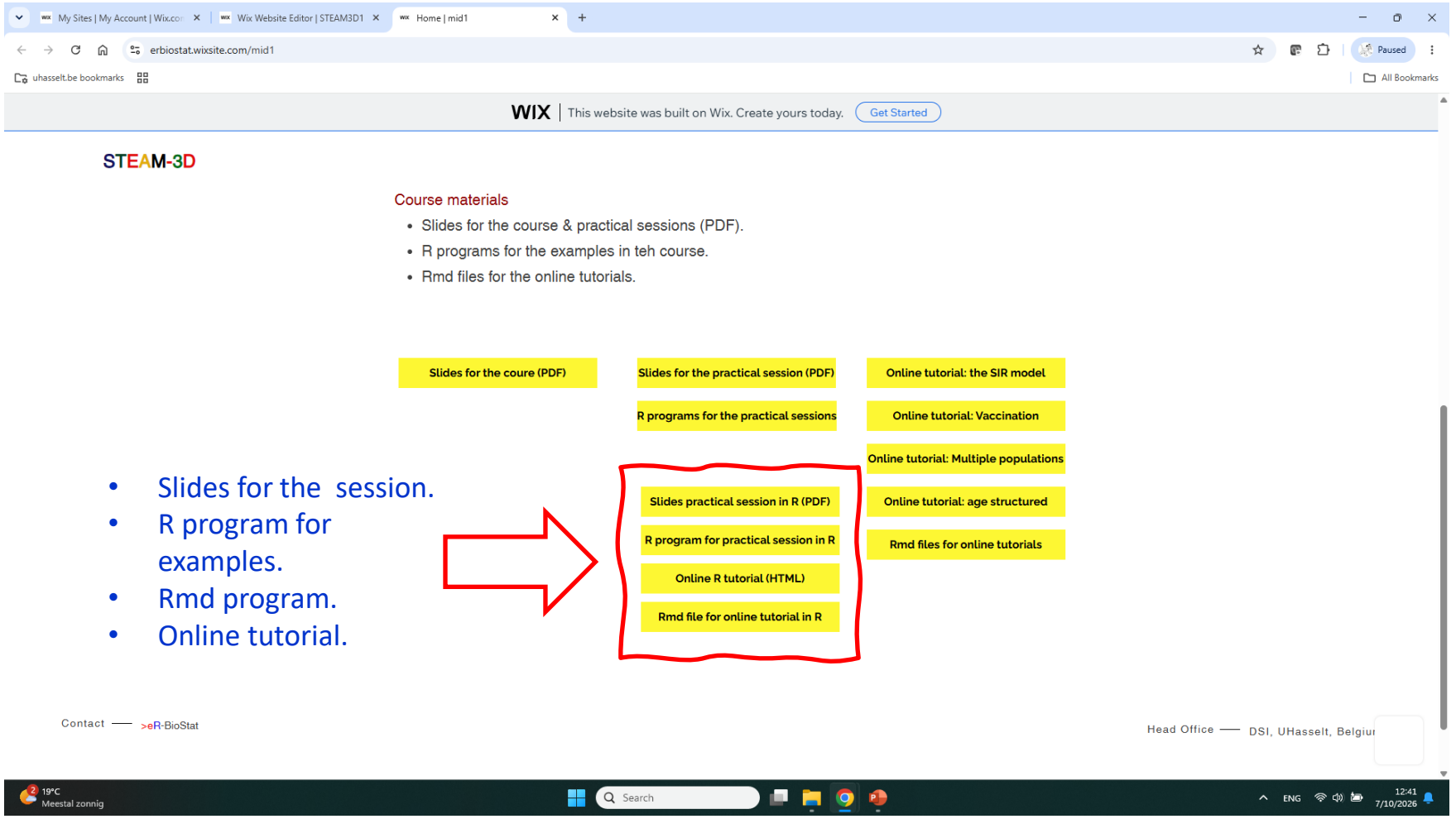
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Practical sessions: R programs



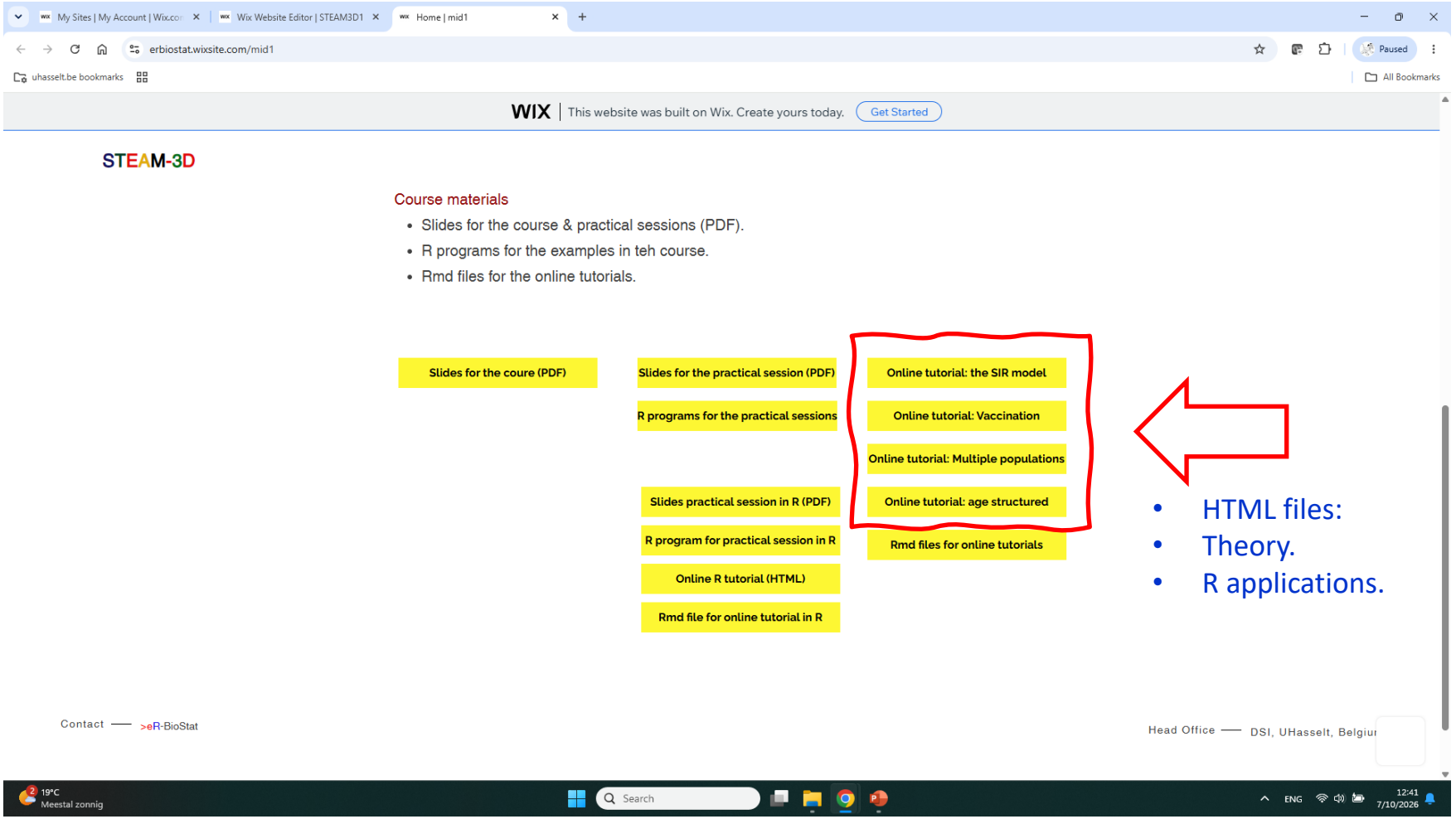
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Introduction session in R

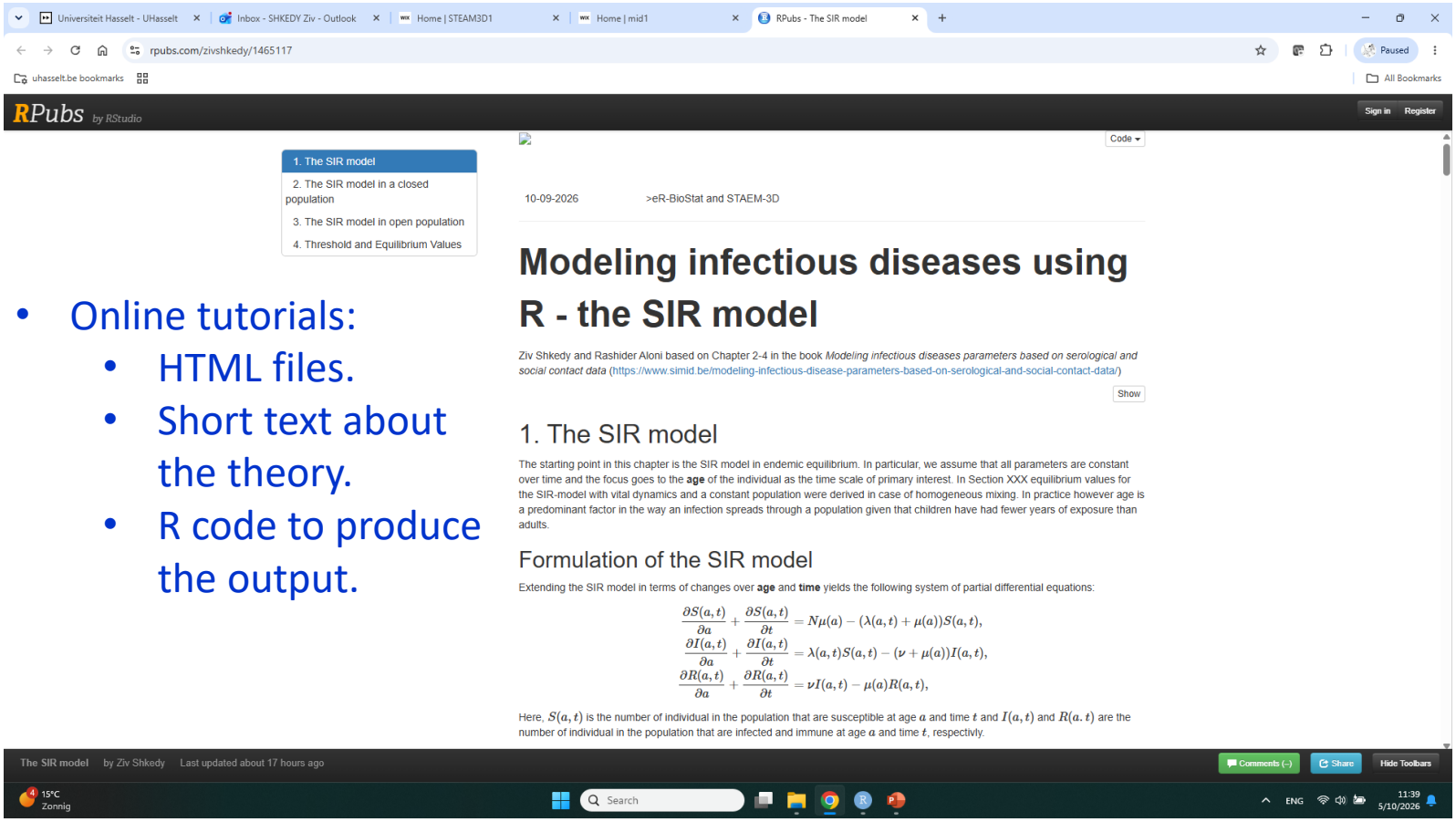


Where to find the course materials ?

Online Tutorials



Where to find the course materials ?
Online tutorials: the HTML files



- Online tutorials:
  - HTML files.
  - Short text about the theory.
  - R code to produce the output.

Where to find the course materials ?

Online Tutorials

